# Numerical Analysis

## Lecture1: Introduction

Instructor: Prof. Qiwei Zhan
Zhejiang University

# Outline


1. Introduction
   - Motivation
   - Objective
2. Computer Arithmetic
3. Algorithms and Convergence

# Motivation

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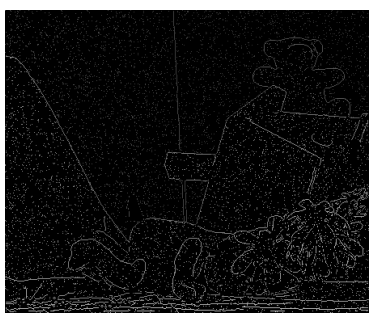
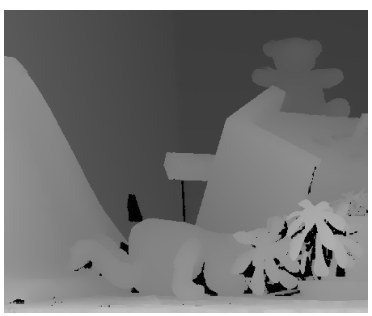
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- Q3: Wireless Channel Capacity

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- Q4:Guided depth upsampling

$$\frac{\partial D(\mathbf{p};t)}{\partial t} = \text{div}\left(\mathbf{W}(\mathbf{p};t)\nabla D(\mathbf{p};t)\right) \quad s.t.\ \ D(\mathbf{p};0) = D_0(\mathbf{p})$$

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  - linear equations
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- Floating point numbers
- Error Analysis
- Floating Point Operations

3 Algorithms and Convergence

# Binary floating point numbers

- short real format (single precision, 4 Bytes, 1 Byte = 8 Bits)
- Long real format (double precision, 8 Bytes)
  - Default data type in MATLAB, 'double' in C
  - Base: 2

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|---|---|---|
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  - http://en.wikipedia.org/wiki/IEEE_floating_point

# Decimal Floating Point Numbers

- Base: 10
- $k$-digit decimal machine number:

$$\pm 0.d_1 d_2 \ldots d_k \times 10^n, \quad 1 \leq d_1 \leq 9, \quad 0 \leq d_i \leq 9$$

- Any positive number within the numerical range can be written:

$$y = 0.d_1 d_2 \ldots d_k d_{k+1} d_{k+2} \ldots \times 10^n$$

- Two ways to represent y with k digits:
  1. Chopping:
  $$fl(y) = 0.d_1 d_2 \ldots d_k \times 10^n$$
  2. Roundoff: Add $5 \times 10^{n-(k+1)}$ and chop:
  $$fl(y) = 0.\delta_1 \delta_2 \ldots \delta_k \times 10^n$$

- Roundoff error

# Errors and Significant Digits

## Errors:

If $p^*$ is an approximation to $p$, the absolute error is $|p - p^*|$, and the relative error is $|p - p^*|/|p|$, provided that $p \neq 0$.

**Ex 2** Determine the absolute and relative errors when approximating $p$ by $p^*$ when

**(a)** $p = 0.3000 \times 10^1$ and $p^* = 0.3100 \times 10^1$;

**(b)** $p = 0.3000 \times 10^{-3}$ and $p^* = 0.3100 \times 10^{-3}$;

**(c)** $p = 0.3000 \times 10^4$ and $p^* = 0.3100 \times 10^4$.

***Solution***

**(a)** For $p = 0.3000 \times 10^1$ and $p^* = 0.3100 \times 10^1$ the absolute error is 0.1, and the relative error is $0.333\overline{3} \times 10^{-1}$.

**(b)** For $p = 0.3000 \times 10^{-3}$ and $p^* = 0.3100 \times 10^{-3}$ the absolute error is $0.1 \times 10^{-4}$, and the relative error is $0.333\overline{3} \times 10^{-1}$.

**(c)** For $p = 0.3000 \times 10^4$ and $p^* = 0.3100 \times 10^4$, the absolute error is $0.1 \times 10^3$, and the relative error is again $0.333\overline{3} \times 10^{-1}$.

# Errors and Significant Digits

## Significant digits:

The number $p^*$ is said to approximate $p$ to $k$ significant digits if $k$ is the largest nonnegative integer for which

$$\frac{|p - p^*|}{|p|} \leq 5 \times 10^{-k}$$

$$\left|\frac{y - fl(y)}{y}\right| = \left|\frac{0.d_1d_2 \ldots d_kd_{k+1} \ldots \times 10^n - 0.d_1d_2 \ldots d_k \times 10^n}{0.d_1d_2 \ldots \times 10^n}\right|$$

$$= \left|\frac{0.d_{k+1}d_{k+2} \ldots \times 10^{n-k}}{0.d_1d_2 \ldots \times 10^n}\right| = \left|\frac{0.d_{k+1}d_{k+2} \ldots}{0.d_1d_2 \ldots}\right| \times 10^{-k}$$

$$\leq \frac{1}{0.1} \times 10^{-k} = 10^{-k+1}$$

# Floating Point Operations

- Floating Point Operations
  - Machine addition: $x \oplus y = fl(fl(x) + fl(y))$
  - Subtraction: $x \ominus y = fl(fl(x) - fl(y))$
  - Multiplication: $x \otimes y = fl(fl(x) \times fl(y))$
  - division: $x \oslash y = fl(fl(x) \div fl(y))$
- Cancelation
  - Common problem: Subtraction of nearly equal numbers:

$$fl(x) = 0.d_1 d_2 \ldots d_p \alpha_{p+1} \alpha_{p+2} \ldots \alpha_k \times 10^n$$

$$fl(y) = 0.d_1 d_2 \ldots d_p \beta_{p+1} \beta_{p+2} \ldots \beta_k \times 10^n$$

gives fewer digits for significance:

$$fl(fl(x) - fl(y)) = 0.\sigma_{p+1} \sigma_{p+2} \ldots \sigma_k \times 10^{n-p}$$

# Floating Point Operations - Errors

Given $x = 5/7$, $u = 0.714251$, $v = 98765.9$, and $w = 0.111111 \times 10^{-4}$.
Determine the five-digit chopping values of

| Operation | Result | Actual value | Absolute error | Relative error |
|---|---|---|---|---|
| $x \ominus u$ | $0.30000 \times 10^{-4}$ | $0.34714 \times 10^{-4}$ | $0.471 \times 10^{-5}$ | 0.136 |
| $(x \ominus u) \oplus w$ | $0.27000 \times 10^{1}$ | $0.31242 \times 10^{1}$ | 0.424 | 0.136 |
| $(x \ominus u) \otimes v$ | $0.29629 \times 10^{1}$ | $0.34285 \times 10^{1}$ | 0.465 | 0.136 |
| $u \oplus v$ | $0.98765 \times 10^{5}$ | $0.98766 \times 10^{5}$ | $0.161 \times 10^{1}$ | $0.163 \times 10^{-4}$ |

# Floating Point Operations - Errors

Consider a quadratic equation $x^2 + 62.10x + 1 = 0$, whose roots are approximately

$$x_1 = -0.01610723 \qquad x_2 = -62.08390.$$

We use four digital rounding arithmetic to find the root.

$$\sqrt{b^2 - 4ac} = 62.06.$$

$$fl(x_1) = \frac{-62.10 + 62.06}{2.000} = -0.02000$$

with the large relative error

$$\frac{|-0.01611 + 0.02000|}{|-0.01611|} \approx 2.4 \times 10^{-1}$$

# Floating Point Operations - Errors

To obtain a more accurate four-digit rounding approximation for $x_1$, we change the quadratic formula into

$$x_1 = \frac{-2c}{b + \sqrt{b^2 - 4ac}}$$

Then

$$fl(x_1) = \frac{-2.000}{62.10 + 62.06} = -0.01610$$

which has the small relative error $6.2 \times 10^{-4}$.

**The lesson: Think before you compute!**

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3 Algorithms and Convergence
- Algorithms
- Convergence

# Algorithms

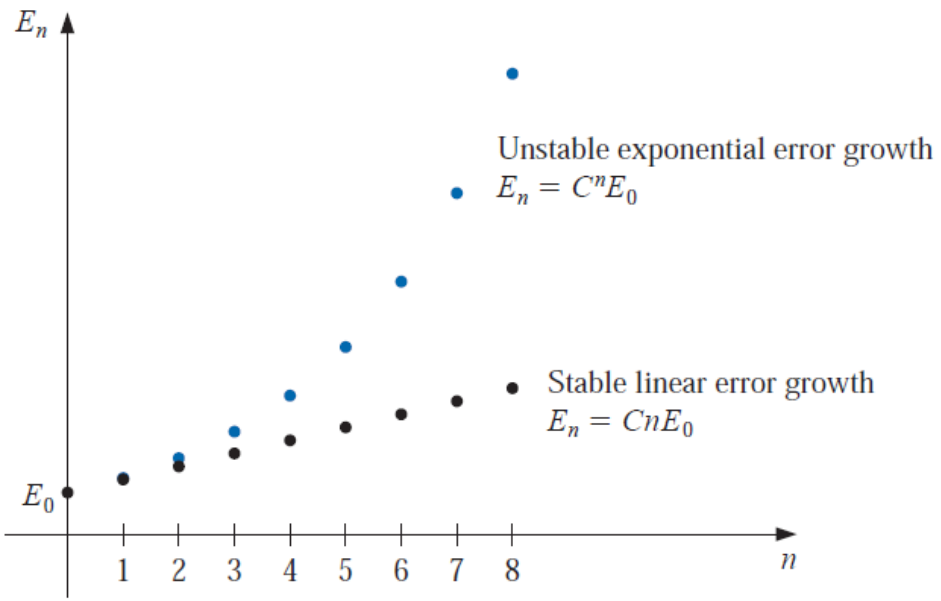
## Definition

An algorithm is a set of operations to sovle a problem or approximate a solution to the problem.

- Growth of error
  Suppose $E_0 > 0$ is an initial error, and $E_n$ is the error after $n$ operations.
  - $E_n \approx CnE_0$: linear growth of error
  - $E_n \approx C^n E_0$: exponential growth of error

# Stability

- Stability properties of algorithms
  - Stable: small changes in the initial data produce small changes in the final result
  - Unstable or conditionally stable: large errors in final results for all or some initial data with small errors



# Rate of convergence (sequences)

## Definition

Suppose $\{\beta_n\}_{n=1}^{\infty}$ is a sequence converging to zero, and $\{\alpha_n\}_{n=1}^{\infty}$ converges to a number $\alpha$. If a positive constant $K$ exists with

$$|\alpha_n - \alpha| \leq K|\beta_n|, \quad \text{for large } n,$$

then we say that $\{\alpha_n\}_{n=1}^{\infty}$ converges to $\alpha$ with rate of convergence $O(\beta_n)$, indicated by $\alpha_n = \alpha + O(\beta_n)$.

## Polynomial rate of convergence

Normally we will use

$$|\beta_n| = \frac{1}{n^p}$$

and look for the largest value $p > 0$ such that $\alpha_n = \alpha + O(\frac{1}{n^p})$

# Rate of convergence (functions)

## Definition

Suppose that $lim_{h\to 0} G(h) = 0$ and $\lim_{h\to 0} F(h) = L$. If a positive constant $K$ exists with

$$|F(h) - L| \leq K|G(h)|, \quad \text{for sufficiently small } h,$$

then we write $F(h) = L + O(G(h))$.

## Polynomial rate of convergence

Normally we will use

$$G(h) = h^p,$$

and look for the largest value $p > 0$ such that $F(h) = L + O(h^p)$.

$\rightarrow$ Take a 'log' operator..., very helpful for finite element method, finite difference method...

# Reading Assignment

- Numerical Analysis, Ch.1, Ch.2.1